Statement of Purpose

Program: M.S. in Electrical and Computer Engineering (Cyber-Physical Systems, Control, and Robotics)
Applicant: Yang Sen (Liam) Lin

In my research accepted to **CoRL 2025**, *"See, Point, Fly,"* I identified a fundamental trade-off in robotic autonomy: geometric rules provide reliability but become brittle in dynamic environments, while end-to-end VLM policies offer adaptability but suffer from hallucinations and high edge latency. My objective is to resolve this tension by architecting **Physics-Aware, Edge-Native Autonomy**—systems that constrain the creativity of VLMs with the rigor of physics and the efficiency of silicon. I specifically seek the **Cyber-Physical Systems** track at UCLA to achieve this, as it uniquely offers the convergence of **Prof. Achuta Kadambi's** computational imaging priors and **Prof. Ankur Mehta's** embedded optimization techniques.

My preparation is defined by the convergence of **Mechanical Engineering** first principles and **Computer Science** algorithmic reasoning. Unlike black-box approaches, my CoRL work formulated a **learning-free** control framework. I utilized **Projective Geometry** to derive non-linear unprojection laws, mapping 2D pixel space directly to **SE(3) velocity primitives**. This formulation solved depth ambiguity without external sensors, optimizing visual tracking complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$. Beyond the technical derivation, I **led the manuscript composition** and delivered a **spotlight presentation** at CoRL 2025, honing my ability to distill complex theoretical concepts for a global audience.

However, I observed that static pin-hole assumptions fail in unstructured environments with complex light transport. The system lacks **differentiable optical priors**—a "Physics Gap" I aim to resolve at UCLA.

Simultaneously, my research philosophy is defined by the constraints of the physical layer. While many focus solely on algorithms, my internship at **Wolley Inc.** grounded me in hardware architecture. I engineered C-based diagnostic tools to profile **CXL Type-3 memory** bandwidth and host pressure, gaining the low-level intuition needed to optimize I/O efficiency for physical AI. I plan to extend this hardware-software co-design mindset during my **upcoming internship at OMRON SINIC X**, where I will distill Robotics Foundation Models for edge deployment. I possess a full-stack perspective: optimizing both the high-level policy and the low-level memory architecture to bridge the "Resource Gap."

UCLA is the ideal environment to synthesize these tracks. To solve "Physical Blindness," I aim to join the **Visual Machines Group**. I propose replacing static heuristics with **learnable optical priors** embedded in VLM attention layers, aligning with the "VLM4D" vision. To address the "Computational Deadlock," I seek **Prof. Ankur Mehta's LEMUR Lab** to co-design hardware-aware compression techniques, enabling heavy foundation models to run on ubiquitous robots.

To support this research, I require a rigorous theoretical upgrade. My background prepares me for **ECE 209AS (Computational Imaging)** to formalize my optical intuition, and **ECE 236B (Convex Optimization)** to transform my heuristic safety checks into provably stable guarantees. My record of zero-shot SOTA results demonstrates methodological velocity; I seek the rigor of the UCLA ecosystem to refine my theoretical depth and define the safety guarantees required for the next generation of Trustworthy Embodied Systems.